Practical No. 11

1. .Display a grid of book covers using GridView for a library catalog.

MainActivity.java :

```
package com.techtitans.madpracticals;

import android.os.Bundle;
import android.widget.GridView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   int[] ids = {
         R.drawable.img_3,
         R.drawable.img_4,
         R.drawable.img_5,
         R.drawable.img_6,
         R.drawable.img_7,
         R.drawable.img_8
   };
   String[] books = {
         "Book 1",
         "Book 2",
         "Book 3",
         "Book 4",
         "Book 5",
         "Book 6",
   };
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      GridView grd = findViewById(R.id.grdView);
      GridAdapter adp = new GridAdapter(this, ids, books);
      grd.setAdapter(adp);
   }
}
```

activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:padding="16dp"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <GridView
      android:id="@+id/grdView"
```

```
    android:numColumns="2"
    android:horizontalSpacing="12dp"
    android:verticalSpacing="12dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent" />
</LinearLayout>
```

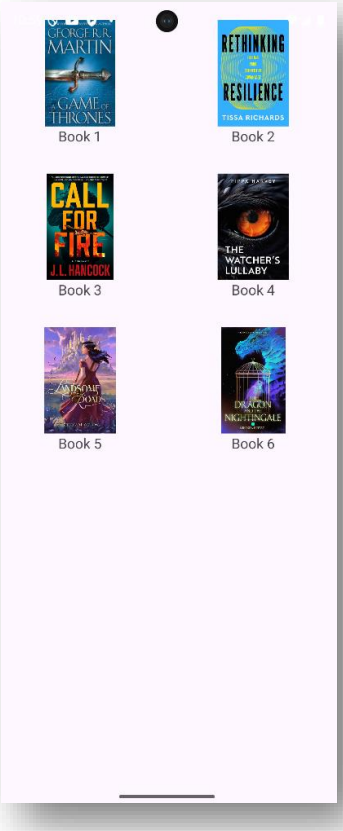
Gridlayout.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:orientation="vertical"
  android:padding="10dp"
  android:layout_width="match_parent"
  android:layout_height="wrap_content">

  <ImageView
    android:id="@+id/imageView"
    android:layout_width="120dp"
    android:layout_height="120dp"
    android:layout_gravity="center"
    android:src="@drawable/img_5" />

  <TextView
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Book"
    android:textSize="16dp"
    android:layout_gravity="center" />
</LinearLayout>
```

GridAdpater:-

```
package com.techtitans.madpracticals;
import android.content.Context;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.BaseAdapter;
import android.widget.ImageView;
import android.widget.TextView;

public class GridAdapter extends BaseAdapter {

  Context c;
  int[] ids;
  String[] b;
```

```
    public GridAdapter(Context c, int[] ids, String[] b) {
        this.c = c;
        this.ids = ids;
        this.b = b;
    }

    public int getCount() {
        return ids.length;
    }

    public Object getItem(int p) {
        return null;
    }

    public long getItemId(int p) {
        return 0;
    }

    public View getView(int p, View v, ViewGroup g) {
        if (v == null) {
            v = LayoutInflater.from(c).inflate(R.layout.gridlayout, g, false);
        }

        ImageView img = v.findViewById(R.id.imageView);
        TextView txt = v.findViewById(R.id.textView2);

        img.setImageResource(ids[p]);
        txt.setText(b[p]);

        return v;
    }
}
```



3.Wrap the entire layout in a ScrollView and test scrolling behavior.
2.Use ImageView to show a banner image for a hotel or library welcome screen.

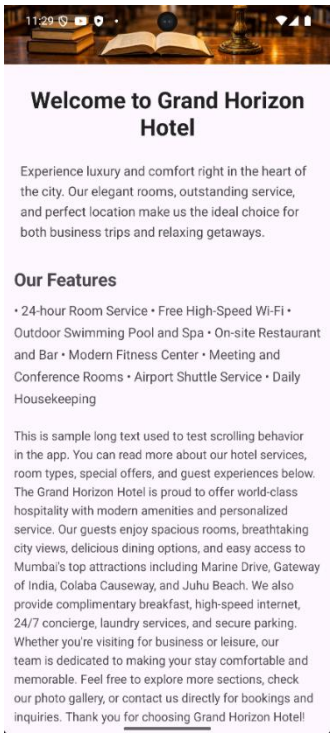
MainActivity.java :

```
package com.techtitans.madpracticals;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:fillViewport="true"
    android:scrollbars="vertical">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical">

        <ImageView
            android:id="@+id/bannerImage"
            android:layout_width="match_parent"
            android:layout_height="220dp"
            android:scaleType="centerCrop"
            android:src="@drawable/img"
            android:contentDescription="@string/welcome_banner"
            android:importantForAccessibility="yes" />

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:padding="24dp"
            android:textSize="28sp"
```

```
        android:textStyle="bold"
        android:textColor="#212121"
        android:gravity="center"
        android:text="@string/welcome_to_our_hotel" />

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:paddingHorizontal="24dp"
        android:paddingBottom="16dp"
        android:textSize="16sp"
        android:lineSpacingExtra="6dp"
        android:textColor="#424242"
        android:text="@string/welcome_description_long" />
i
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="@string/our_features"
        android:textSize="22sp"
        android:textStyle="bold"
        android:paddingBottom="12dp"/>

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="@string/features_list"
        android:textSize="16sp"
        android:lineSpacingExtra="8dp"
        android:paddingBottom="24dp"/>

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="@string/lorem_ipsum_long"
            android:textSize="15sp"
            android:lineSpacingExtra="5dp"/>

    </LinearLayout>
  </LinearLayout>
</ScrollView>
```



Question 4 Create a ListView showing recent borrowers or hotel guests with names and check-in dates.
Question 6 Add click listeners to ListView items to show detailed information in a Toast.

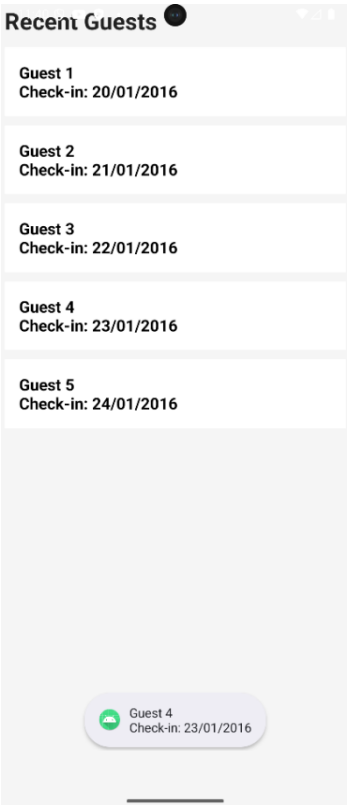
MainActivity2.java :

```
package com.techtitans.madpracticals;

import android.os.Bundle;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity2 extends AppCompatActivity {

   String[] guests = {
         "Guest 1\nCheck-in: 20/01/2016",
         "Guest 2\nCheck-in: 21/01/2016",
         "Guest 3\nCheck-in: 22/01/2016",
         "Guest 4\nCheck-in: 23/01/2016",
         "Guest 5\nCheck-in: 24/01/2016"
   };

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main2);

      ListView lst = findViewById(R.id.lsttview);

      ArrayAdapter adp = new ArrayAdapter(
            this,
            R.layout.list_item,
            R.id.textView,
            guests
      );

      lst.setAdapter(adp);

      lst.setOnItemClickListener(new AdapterView.OnItemClickListener() {
         @Override
         public void onItemClick(AdapterView<?> parent, android.view.View view, int position,
long id) {
            Toast.makeText(
```

```
                MainActivity2.this,
                guests[position],
                Toast.LENGTH_SHORT
            ).show();
        }
    });
  }
}
```

activity_main2.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:background="#F5F5F5"
  android:padding="16dp">

  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Recent Guests"
    android:textSize="26sp"
    android:textStyle="bold"
    android:textColor="#212121"
    android:layout_marginBottom="12dp" />

  <ListView
    android:id="@+id/lsttview"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:dividerHeight="10dp"
    android:divider="@android:color/transparent" />

</LinearLayout>
```



List_item.xml :-

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com
/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="wrap_content"
  android:background="#FFFFFF"
  android:padding="16dp"
  android:orientation="vertical">
```

```
    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:textStyle="bold"
        android:textColor="#000000" />

</LinearLayout>
```

Q5. Implement a custom adapter to show book titles and images in GridView.
Q7. Combine GridView and ListView in a single screen to simulate a dashboard.

MainActivity.java :

```
package com.techtitans.madpracticals;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;


import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.GridView;
import android.widget.ListView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    int[] ids = {
            R.drawable.img,
            R.drawable.img_4,
            R.drawable.img_6,
            R.drawable.img_3
    };

    String[] books = {
            "Book One",
            "Book Two",
            "Book Three",
            "Book Four"
    };

    String[] guests = {
            "Guest 1\nCheck-in: 20/01/2016",
            "Guest 2\nCheck-in: 21/01/2016",
```

```
        "Guest 3\nCheck-in: 22/01/2016",
        "Guest 4\nCheck-in: 23/01/2016"
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        GridView grd = findViewById(R.id.grdView);
        ListView lst = findViewById(R.id.lstview);

        GridAdapter gadp = new GridAdapter(this, ids, books);
        grd.setAdapter(gadp);

        ArrayAdapter ladp = new ArrayAdapter(this, R.layout.list_item, R.id.textView, guests);
        lst.setAdapter(ladp);
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:fillViewport="true">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:background="#F5F5F5"
        android:padding="16dp">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Library Dashboard"
            android:textSize="26sp"
            android:textStyle="bold"
            android:textColor="#212121"
            android:layout_marginBottom="12dp" />

        <GridView
            android:id="@+id/grdView"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
```

```
        android:numColumns="2"
        android:horizontalSpacing="12dp"
        android:verticalSpacing="12dp"
        android:stretchMode="columnWidth"
        android:layout_marginBottom="20dp" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Recent Guests"
        android:textSize="22sp"
        android:textStyle="bold"
        android:textColor="#212121"
        android:layout_marginBottom="8dp" />

    <ListView
        android:id="@+id/lstview"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:divider="@android:color/transparent"
        android:dividerHeight="10dp" />

  </LinearLayout>
</ScrollView>
```

List_item.xml

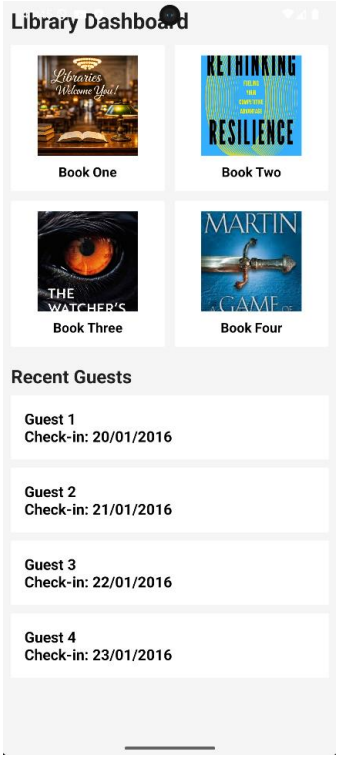
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="wrap_content"
  android:orientation="vertical"
  android:background="#FFFFFF"
  android:padding="16dp">

  <TextView
    android:id="@+id/textView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textSize="18sp"
    android:textStyle="bold"
    android:textColor="#000000" />

</LinearLayout>
```

grid_item :-

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:background="#FFFFFF"
    android:padding="16dp">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:textStyle="bold"
        android:textColor="#000000" />

</LinearLayout>
```

GridAdapter.java:-

```
package com.techtitans.madpracticals;


import android.content.Context;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.BaseAdapter;
import android.widget.ImageView;
import android.widget.TextView;

public class GridAdapter extends BaseAdapter {

    Context c;
    int[] ids;
    String[] t;

    public GridAdapter(Context c, int[] ids, String[] t) {
        this.c = c;
        this.ids = ids;
        this.t = t;
    }

    public int getCount() {
        return ids.length;
    }

    public Object getItem(int p) {
        return null;
    }
```

```
    public long getItemId(int p) {
        return 0;
    }

    public View getView(int p, View v, ViewGroup g) {
        if (v == null) {
            v = LayoutInflater.from(c).inflate(R.layout.grid_item, g, false);
        }

        ImageView img = v.findViewById(R.id.imageView);
        TextView txt = v.findViewById(R.id.textView2);

        img.setImageResource(ids[p]);
        txt.setText(t[p]);

        return v;
    }
}
```

output :-